Product Requirements Document (PRD)

NeuroTriage AI

AI-assisted Emergency Stroke Triage Platform

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Revision History

Version	Date	Author	Description
1.0	05-Aug-2026	Saksha Hegde	Initial Product Requirements Document

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1. Document Overview

Purpose

This document defines the functional and non-functional requirements for **NeuroTriage AI**, an AI-assisted emergency stroke triage platform.

It serves as the primary specification for engineering, UX, AI, and clinical teams responsible for building and validating the MVP.

The objective of this document is to ensure that all stakeholders have a common understanding of:

- the problem being solved,
- expected product behaviour,
- user workflow,
- business rules,
- product scope,
- and acceptance criteria.

Product Scope

The MVP focuses on assisting radiologists in prioritizing suspected intracranial hemorrhage (ICH) cases immediately after CT brain acquisition.

The product provides:

- AI-assisted worklist prioritization
- AI confidence scoring
- AI-assisted reading support
- Radiologist feedback capture

The product does **not** replace clinical diagnosis.

Intended Audience

This document is intended for:

- Product Management
- Software Engineering
- AI Engineering
- UX Design
- Clinical Validation Team
- Project Stakeholders

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2. Product Principles

Every design and engineering decision shall align with the following product principles.

Principle	Description
AI Assists, Never Replaces	AI supports prioritization while the radiologist remains responsible for diagnosis.
Workflow Before Technology	The product integrates into existing PACS workflows instead of introducing new workflows.
Patient Safety First	When uncertainty exists, the product favours earlier human review rather than increased automation.
Transparency Builds Trust	Every AI recommendation includes confidence and explainability information.
Deliver Clinical Value	Product success is measured by improved workflow efficiency and patient outcomes rather than AI accuracy alone.

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3. Product Overview

NeuroTriage AI is an AI-assisted clinical decision support product that intelligently prioritizes emergency CT brain studies within the radiologist's existing PACS worklist.

Rather than replacing the radiologist, the product continuously analyses newly acquired CT studies and assists clinicians by highlighting suspected intracranial hemorrhage cases that require earlier review.

The product is organised into three logical subsystems.

Subsystem 1 – AI Triage Engine

Responsible for:

- Receiving newly acquired CT studies
- Performing AI inference
- Predicting suspected intracranial hemorrhage
- Calculating AI confidence
- Assigning study priority

This subsystem operates entirely in the background without requiring user interaction.

Subsystem 2 – PACS Worklist Experience

Responsible for:

- Displaying incoming CT studies
- Showing AI processing status
- Automatically reprioritizing completed studies
- Presenting clear visual priority indicators

This subsystem delivers the primary workflow benefit of the product.

Subsystem 3 – Reading Experience

Responsible for:

- Displaying CT images
- Presenting AI findings
- Displaying confidence information
- Showing explainability overlays
- Capturing radiologist feedback

This subsystem supports informed clinical decision-making while maintaining clinician control.

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4. Product Goals

Goal Category	Objective
Clinical	Reduce the time required to identify suspected intracranial hemorrhage cases.
Workflow	Integrate AI seamlessly into the existing PACS workflow with minimal additional user interaction.
Trust	Increase clinician confidence through transparent AI recommendations and explainability.
Business	Deliver measurable workflow improvements while providing a scalable enterprise solution.

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5. Success Metrics

The success of NeuroTriage AI will be measured using both product and technical outcomes.

Category	Success Metric
Clinical	Reduced Time-to-Review for suspected ICH cases
Technical	High Sensitivity with low False Negative Rate
Operational	Low AI inference time
User	High Radiologist Adoption Rate
Trust	Low Radiologist Override Rate
Business	Successful pilot deployment and enterprise adoption

Detailed launch thresholds are defined in the **Data, Model, and Evaluation Strategy** document.

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6. Stakeholders

Stakeholder	Responsibility
Radiologist	Reviews prioritized studies and makes the final diagnosis.
Emergency Physician	Uses earlier radiology findings to support patient management.
Hospital Administrator	Evaluates workflow efficiency and operational value.
Product Manager	Owns product vision, roadmap, and feature prioritization.
AI Engineering Team	Develops and maintains AI models.
Software Engineering Team	Develops and maintains the application.
Clinical Validation Team	Validates product performance before deployment.

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7. User Personas

Primary Persona – Radiologist

Primary Goal

Review emergency CT studies as early as possible without increasing workload.

Pain Points

- High imaging volumes.
- Difficulty identifying urgent cases.
- Time pressure during reporting.
- Limited trust in opaque AI systems.

Secondary Persona – Emergency Physician

Primary Goal

Receive imaging results earlier to support timely treatment decisions.

Tertiary Persona – Hospital Administrator

Primary Goal

Improve emergency workflow efficiency while ensuring safe and sustainable adoption of AI technologies.

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8. Functional Requirements and Product Behaviour

The NeuroTriage AI MVP consists of four primary product behaviours that together support the emergency radiology workflow.

1. AI Triage Processing
2. PACS Worklist Experience
3. Reading Experience
4. Feedback Capture

The workflow begins immediately after a CT Brain study is acquired and ends when the radiologist confirms or overrides the AI recommendation.

8.1 End-to-End Workflow

The product shall support the following workflow:

1. A new CT Brain study is acquired.
2. The study is automatically sent to the AI Triage Engine.
3. AI analyses the complete CT study.
4. The AI predicts whether Intracranial Hemorrhage (ICH) is suspected.
5. A confidence score is generated.
6. The study priority is automatically updated.
7. The worklist refreshes without user intervention.
8. The radiologist opens the study.
9. AI findings are displayed alongside the CT images.
10. The radiologist reviews the study and records the final clinical decision.
11. Radiologist feedback is stored for future model improvement.

8.2 AI Triage Processing

Functional Requirements

ID	Requirement	Priority
AI-01	The system shall automatically analyse every newly acquired CT Brain study without user intervention.	Must
AI-02	The system shall predict whether intracranial hemorrhage is suspected.	Must
AI-03	The system shall generate a confidence score for every prediction.	Must
AI-04	The system shall assign a worklist priority based on the prediction and confidence level.	Must
AI-05	The system shall complete analysis before updating the worklist.	Must

Business Rules

- AI processing begins automatically after image acquisition.
- Every study receives exactly one prediction.
- Every prediction includes a confidence score.
- AI does not make the clinical diagnosis.
- Priority assignment follows the product prioritization policy.

Acceptance Criteria

- AI processing starts automatically for every new study.
- Every processed study receives a prediction.
- Every prediction includes a confidence value.
- Worklist priority updates only after AI processing is complete.

8.3 PACS Worklist Experience

Functional Requirements

ID	Requirement	Priority
WL-01	The worklist shall display all available CT Brain studies.	Must
WL-02	The worklist shall display AI processing status for newly received studies.	Must
WL-03	The worklist shall automatically reorder studies after AI processing completes.	Must
WL-04	The worklist shall display a visual priority indicator for every completed study.	Must
WL-05	Users shall be able to open any study directly from the worklist.	Must

Business Rules

- Newly received studies initially display an **AI Processing** status.
- Priority updates occur automatically.
- High-priority studies appear above lower-priority studies.
- Manual sorting is outside the MVP scope.

Acceptance Criteria

- New studies appear immediately after acquisition.
- Worklist updates automatically without requiring page refresh.
- High-priority studies move to the correct position.
- Selecting a study opens the Reading Experience.

8.4 Reading Experience

Functional Requirements

ID	Requirement	Priority

RD-01	The system shall display the CT study selected from the worklist.	Must
RD-02	The system shall display the AI prediction.	Must
RD-03	The system shall display the AI confidence score.	Must
RD-04	The system shall display explainability information highlighting suspected regions.	Should
RD-05	The radiologist shall be able to confirm or override the AI recommendation.	Must

Business Rules

- AI findings are presented as decision support only.
- Confidence information is always displayed together with the prediction.
- Explainability information supports interpretation but is not mandatory for diagnosis.
- Final diagnosis remains the responsibility of the radiologist.

Acceptance Criteria

- The selected CT study loads successfully.
- AI prediction and confidence are visible.
- Radiologist actions are available.
- Override actions do not require restarting AI analysis.

8.5 Feedback Capture

Functional Requirements

ID	Requirement	Priority
FB-01	The system shall record the radiologist’s final decision.	Must

FB-02	The system shall record AI predictions for comparison.	Must
FB-03	The system shall store override actions for future analysis.	Should
FB-04	The system shall support future model improvement using captured feedback.	Could

Business Rules

- Feedback collection does not interrupt clinical workflow.
- Feedback is associated with the corresponding AI prediction.
- Captured feedback is used for continuous product improvement.

Acceptance Criteria

- Radiologist feedback is successfully recorded.
- AI prediction and final decision are linked.
- Override actions are available for later review.

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9. Non-Functional Requirements

The following non-functional requirements define the expected quality attributes of NeuroTriage AI.

Category	Requirement
Performance	AI analysis should complete within the target inference time defined in the Evaluation Strategy.
Reliability	The product shall continue to function reliably during continuous emergency department operation.
Availability	The system should remain available throughout routine clinical operations with minimal downtime.
Usability	AI recommendations shall be presented in a simple, intuitive manner requiring minimal additional user interaction.

Explainability	Every AI prediction shall be accompanied by confidence information and explainability support.
Security	Patient information shall be protected through anonymization and secure data handling practices.
Scalability	The product shall support deployment across multiple hospitals without requiring changes to core functionality.
Maintainability	AI models and software components shall support future updates without disrupting existing workflows.

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10. User Stories

US-01

As a Radiologist
I want newly acquired CT brain studies to be analysed automatically
So that emergency cases are identified without manual intervention.

US-02

As a Radiologist
I want suspected intracranial hemorrhage cases to appear higher in my worklist
So that I can review critical patients earlier.

US-03

As a Radiologist
I want every AI prediction to include a confidence score
So that I can better judge the reliability of the recommendation.

US-04

As a Radiologist
I want to view AI explainability information alongside the CT images
So that I understand why the AI reached its prediction.

US-05

As a Radiologist
I want to confirm or override the AI recommendation
So that final clinical responsibility remains under my control.

US-06

As a Hospital Administrator
I want the product to integrate with existing radiology workflows
So that productivity improves without disrupting current clinical practice.

US-07

As a Product Manager
I want radiologist feedback to be captured
So that future versions of the AI can continuously improve.

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11. MVP Scope

The MVP demonstrates the core value proposition of NeuroTriage AI by simulating an AI-assisted emergency radiology workflow.

Included

- PACS worklist simulation
- Automatic AI processing of incoming CT Brain studies
- AI-assisted worklist prioritization
- Reading screen with CT image viewer
- AI prediction and confidence display
- Explainability visualization
- Radiologist confirmation or override
- Feedback capture

Excluded

The following capabilities are intentionally excluded from the MVP:

- Hospital authentication and user management
- Live PACS integration
- DICOM networking
- Multi-user collaboration
- Clinical report generation
- Regulatory workflows
- Production deployment infrastructure

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12. Assumptions and Dependencies

Assumptions

- CT Brain studies are available immediately after image acquisition.
- AI inference is completed before worklist prioritization.
- Radiologists remain the final decision makers.
- The AI model has been clinically validated before deployment.

Dependencies

The successful implementation of NeuroTriage AI depends on:

- Availability of anonymized CT Brain datasets
- AI inference service
- Existing PACS workflow
- Clinical validation by expert radiologists
- Product approval by hospital stakeholders

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13. Product Risks

Risk	Potential Impact	Mitigation
Missed ICH (False Negative)	Delayed diagnosis of critical patients	Human-in-the-Loop review and conservative prioritization policy

Over-reliance on AI	Reduced clinical vigilance	Confidence scores, explainability, and clinician override
Workflow Disruption	Reduced clinician adoption	Pilot deployment and gradual rollout
Model Performance Drift	Reduced long-term accuracy	Continuous monitoring and periodic model updates

Detailed governance controls are documented in the **Ethics, Governance, and Risk Note**.

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14. Future Enhancements

The MVP establishes the foundation for AI-assisted emergency stroke triage. Future product releases will expand the platform while preserving the core product principles of patient safety, clinician trust, and seamless workflow integration.

Phase 2 – Clinical Expansion

Future enhancements may include:

- Support for additional neurological emergencies such as ischemic stroke and brain tumors.
- Enhanced explainability visualizations to improve clinician confidence.
- Feedback-driven model improvement using validated clinical outcomes.
- Workflow analytics dashboard for monitoring operational efficiency.

Phase 3 – Enterprise Platform

Long-term product evolution may include:

- Multi-condition AI detection within a single workflow.
- AI-assisted preliminary reporting to support radiologist productivity.
- Multi-hospital deployment with centralized model management.
- Continuous performance monitoring and governance dashboards.

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15. Requirements Traceability Matrix

The following matrix demonstrates how the major product requirements map to business objectives and user-facing functionality.

Requirement	Product Objective	Prototype Component
Automatic AI analysis of incoming CT studies	Reduce time to identify emergency cases	AI Processing
Dynamic worklist prioritization	Improve emergency workflow efficiency	PACS Worklist
AI confidence display	Increase clinician trust	Reading Screen
Explainability support	Improve transparency of AI recommendations	Reading Screen
Radiologist confirmation and override	Maintain clinician control	Reading Screen
Feedback capture	Support continuous product improvement	Feedback Module

This traceability ensures that every major product capability contributes directly to the overall business and clinical objectives.

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16. Glossary

Term	Description
AI	Artificial Intelligence
ICH	Intracranial Hemorrhage
PACS	Picture Archiving and Communication System
CT	Computed Tomography
ROI	Region of Interest

MVP	Minimum Viable Product
Human-in-the-Loop	Workflow in which the radiologist remains responsible for the final clinical decision while AI provides decision support.

17. Document Approval

This Product Requirements Document represents the agreed product scope and behaviour for Version 1.0 of NeuroTriage AI.

Any future changes to product functionality should be evaluated against the product principles defined in this document to ensure continued alignment with the overall vision, user needs, and business objectives.

End of Document